

ITEM: {{ QUESTION_ID: RX-C2-04 }}

DEMAND: {{ DIFFICULTY_BADGE: STANDARD }}

TARGET GRADE: GRADE 9 ADVANCED

PRIMARY: {{ PRIMARY_CONCEPT: RX-S03 Oxidising vs Reducing Agents }}

SUPPORTS: {{ SUPPORTING_CONCEPTS: [RX-S01, RX-S02] }}

SRC: {{ SOURCE: JEE MAIN 2022 - 27 JUL SHIFT 1 }}

{{ QUESTION_TITLE: Reducing Agent Identification & Electron Transfer in Acidic Permanganate-Oxalate Reaction }}

ORIGINAL EXAMSIDE ITEM

{{ QUESTION_STEM }}:

In the acidic aqueous redox reaction:

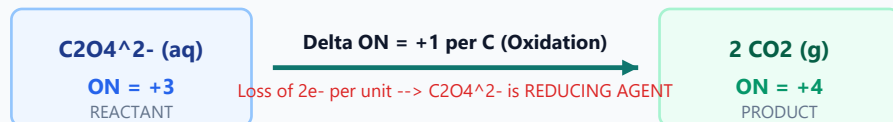


identify the reducing agent, the oxidation state change of the reducing element, and the total moles of electrons transferred per mole of reducing agent.

Options:

- (a) MnO_4^- , Mn changes from +7 to +2, 5 electrons
- (b) $\text{C}_2\text{O}_4^{2-}$, C changes from +4 to +3, 1 electron
- (c) $\text{C}_2\text{O}_4^{2-}$, C changes from +3 to +4, 2 electrons
- (d) H^+ , H changes from +1 to 0, 1 electron

{{ VISUAL_PRIMITIVE: Carbon Oxidation & Role Assignment Lane }}



H1-H3 HINT LADDER (READ ONLY IF STUCK)

* {{ H1_NOTICE }}: Assign oxidation numbers to Carbon in reactant $\text{C}_2\text{O}_4^{2-}$ and product CO_2 .

* {{ H2_MODEL }}: The reducing agent is the REACTANT that is OXIDISED (increases ON).

Calculate ON jump per Carbon atom, then multiply by 2 Carbon atoms per $\text{C}_2\text{O}_4^{2-}$.

* {{ H3_START }}: In $\text{C}_2\text{O}_4^{2-}$: $2\text{C} + 4(-2) = -2 \Rightarrow 2\text{C} = +6 \Rightarrow \text{C} = +3$.

In CO_2 : $\text{C} + 2(-2) = 0 \Rightarrow \text{C} = +4$. $\Delta \text{ON} = +1$ per C. Two carbons $\Rightarrow 2$ electrons lost.

CHEMISTRY REASONING ROUTE

Step 1 [PARSE & BIND]: Identify reactants and spectator ions. H^+ acts as medium carrier.

Step 2 [TRACK ON]: Mn in MnO_4^- (+7) $\rightarrow$ Mn^{2+} (+2) [Gain of $5e^-$, Reduction].

Step 3 [TRACK REDUCTANT]: C in $\text{C}_2\text{O}_4^{2-}$ (+3) $\rightarrow$ C in CO_2 (+4) [Loss of $1e^-$ per C, Oxidation].

Step 4 [ATTACH ROLE]: Oxalate ion $\text{C}_2\text{O}_4^{2-}$ undergoes oxidation $\Rightarrow$ It is the REDUCING AGENT.

Step 5 [COUNT ELECTRONS]: 1 formula unit of $\text{C}_2\text{O}_4^{2-}$ contains 2 C atoms $\Rightarrow 2 \times 1e^- = 2e^-$ transferred.

VERIFIED SOLUTION & SYSTEM VERIFICATION

{{ SOLUTION_FINAL_ANSWER }}: Correct Option is (c).

Chemical Language Response: The reducing agent is oxalate ion $\text{C}_2\text{O}_4^{2-}$.

Carbon changes from oxidation state +3 to +4, releasing 2 moles of electrons per mole of $\text{C}_2\text{O}_4^{2-}$.

{{ VERIFICATION_ROUTE }}:

1. Total Electron Balance Check:

Total electrons lost by 5 moles of $\text{C}_2\text{O}_4^{2-} = 5 \times 2e^- = 10$ moles of electrons.

Total electrons gained by 2 moles of $\text{MnO}_4^- = 2 \times 5e^- = 10$ moles of electrons.

Electrons lost = Electrons gained = $10e^-$. [100% ELECTRON CONSERVATION PASS]

2. Atom Conservation Check: 2 Mn, 10 C, 24 O, 16 H on both sides. [ATOM CONSERVATION PASS]

3. Charge Conservation Check: $2(-1) + 5(-2) + 16(+1) = +4 = 2(+2)$. [CHARGE BALANCE PASS]

4. {{ CONDITION_EXCEPTION_CHECK }}: Reaction is recorded in acidic solution; product is Mn^{2+} . [PASS]

Linkage: Traceable to Core1 Lesson RX-ST03 (Agent Roles & Electron Conservation).

{{ H0_ATTEMPT_WORKSPACE: Work Here * Try Before Looking at Hints }}

Your Attempt: Option [____] Oxidant: _____ Reductant: _____